

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0859

West Canton 230 kV
April 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0859. The Interconnection Customer proposes a 227.9 MW wind facility interconnecting at the West Canton 230 kV substation in the Ohio area. The request was received on September 06, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 303 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$6,023,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

W CANTON to E AKRON 230.0 kV line: 107.4% loading (659 MVA) following the outage of the WEST CANTON to NORTH CAMBRIDGE 230.0 kV; DFAX = 37.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W CANTON SUBSTATION to E AKRON 230.0 kV	Reconductor line	\$8,227,000	73% (\$6,023,000)
	Total allocated to project		\$6,023,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.